LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Familiarization with various parts of aircraft and different arrangement of components as seen on various aircraft in station.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Familiarization with various parts of aircraft and different arrangement of components as seen on various aircraft in station.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of structural elements on station aircraft; Safety: Maintain safe distance from rotating parts; Supervision: Instructor monitors trainee proximity to engine and high-voltage zones.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary functions of an aircraft structure?
(b) Name one major structural element found on the fuselage.
(c) Why is the arrangement of components critical for aircraft operation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fuselage structural elements and configuration • Wing assembly components and attachment points • Empennage parts and control surface arrangement • Landing gear types and retraction mechanisms • Engine mounting and nacelle integration	L & Ds	PPT, IB	15 min
(b)	• Comparison of component layouts across different aircraft types • Identification of access panels and maintenance points • Review of structural load paths and stress distribution • Analysis of cockpit instrumentation placement logic • Examination of fuel and hydraulic system routing	L & Ds	PPT, PDF	20 min
(c)	• Detailed inspection of primary structural frames and ribs • Verification of secondary structure and skin panels • Assessment of corrosion protection and sealing methods • Understanding of composite vs metallic structural usage • Documentation of observed component variations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the fuselage in maintaining aircraft structural integrity.
(b) Describe how the arrangement of wings affects the aircraft's lift generation.
(c) Identify two key differences in component arrangement between the aircraft currently in station and a previous model.

12. Home Assignment and Follow Up	Prepare a brief report comparing the structural layout of the aircraft in station with one other aircraft type, highlighting three distinct differences in component arrangement.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Overall consideration of aircraft loading action.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min